



MATHEMATICS

Linear Models Homework and Exit Ticket

Grade 8 | Homework

Build linear models from tables and contexts and interpret slope and initial value with units.

INSIDE THIS RESOURCE

Teacher guidance • 16 practice prompts • Reflection exit ticket • Complete answer guidance

Print the practice pages separately from the teacher key. Provide extra paper for extended work.



Teacher guide

NC.8.F.4 - Analyze functions that model linear relationships; determine and interpret rate of change and initial value from two values or from a graph, table, or description.

Suggested time: 40-50 minutes

Learning focus: Build linear models from tables and contexts and interpret slope and initial value with units.

Teaching sequence: Review prior instruction and model the first prompt with its answer. Assign one four-prompt page at a time. Ask for a check, a reason, or a specific example. Use the reflection ticket to plan the next lesson; it is not a standardized mastery score.

Model prompt: Find slope through (2,7) and (6,19).

Model answer: 3

Support options

AIG: justify a second method or create a transfer example. 504: follow the learner's plan for time, breaks, format, and reduced copying. EC: model steps and accept aligned oral, drawn, or written responses. ESL: preview vocabulary, use bilingual references, and rehearse before writing.

Source and scope: North Carolina Department of Public Instruction, Mathematics Standards Vertical Progression: <https://www.dpi.nc.gov/documents/publications/catalog/ma196-vertical-progression-math/open>; NC Check-Ins 2.0

Mathematics Specifications:

<https://www.dpi.nc.gov/documents/accountability/testing/nc-checkin/nc-check-ins-20-mathematics-grades-3-8-specifications> (accessed September 14, 2026).



Practice 1–4

Show work or explain your response. Use additional paper when needed. Teachers may cut apart the four bordered prompts.

1. Find slope through (2,7) and (6,19).

2. Write $y=mx+b$ when $m=4$ and $b=-2$.

3. Table: x 0,2,4; y 5,11,17. Find m and b.

4. A taxi charges \$4 plus \$2.50 per mile. Model.



Practice 5–8

Show work or explain your response. Use additional paper when needed. Teachers may cut apart the four bordered prompts.

5. Interpret 4 in the taxi model.

6. Find taxi cost for 8 miles.

7. Line through (3,10) with slope 2. Find b .

8. Compare $y=3x+1$ and $y=3x-5$.



Practice 9–12

Show work or explain your response. Use additional paper when needed. Teachers may cut apart the four bordered prompts.

9. A tank has 90 L and drains 6 L/min. Model.

10. When is tank at 42 L?

11. Does (5,16) lie on $y=3x+1$?

12. From y values 8,13,18 as x rises by 1, find rate.



Practice 13–16

Show work or explain your response. Use additional paper when needed. Teachers may cut apart the four bordered prompts.

13. Write context for $y=12x+30$.

14. Explain negative slope in context.

15. Find equation through $(0,-3),(4,5)$.

16. State one limitation of a linear model.



Reflection exit ticket

Complete independently. Refer to your completed practice, but explain in your own words. This checks reflection and application, not a test score prediction.

1. State the focus of this resource in your own words.
2. Create one accurate example connected to the practice.
3. Explain how you checked that example.
4. Identify one possible error and show a correction.
5. Name one specific next practice step and explain why it will help.



Teacher answer guidance

1. 3
2. $y=4x-2$
3. $m=3$, $b=5$
4. $C=2.5m+4$
5. \$4 initial fee
6. \$24
7. 4
8. Same rate; first starts 6 units higher
9. $V=90-6t$
10. 8 minutes
11. Yes
12. 5
13. Answers vary; \$30 start plus \$12 per unit
14. Output decreases as input increases
15. $y=2x-3$
16. Constant rate may not continue outside observed range.

Reflection criteria

Items 1–4: 2 points for an accurate, specific response; 1 for a partly correct response; 0 for a missing or unrelated response. Accept justified alternatives. Item 5 is an unscored planning reflection. Feedback should identify a next teaching move, not label a learner.

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